Lock-Lattice Phase Memory

A Physical Realizability Test for the Lock Law of
The Actualization Theory (AT)

Proposal — ResearchY-NP_170 (rev. 2)

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https://github.com/MagusDraconis/AT

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Abstract

The Actualization Theory (AT) — a reconstruction of physics from Difference, Actualization and Spectrum, in which every observable is derived from the D96 spectral structure while the Standard Model Lagrangian is hosted rather than derived — supports its lock law (Ch8; QG307–QG319), at present, entirely by statistical evidence: datasets and synthetic cohorts. This document specifies an experiment that would test it as a *dynamical* claim, using an engineered nonlinear oscillator ring whose coupling graph is the canonical circulant $C_{96}(1..6)$. The proposal deliberately abandons the energy-storage framing, which is refuted on density grounds, and identifies one idea that survives: the lock lattice as a *phase-memory* and *threshold* element whose distinguishing feature is a *predicted* criticality rather than a fitted one. Three reasons justify the experiment (theory falsification; phase memory where density is irrelevant; a neuromorphic threshold element with predicted sharpness), and the exact procedure — platform, protocol, metrics, blinding, pre-registration and kill criteria — is given in full. Result revision 2 incorporates the results that landed after the first draft (NP_169 and T_010–T_015): the retention controls are now matched on the degeneracy structure, the D96 lattice is presented as the derivable but low-capacity choice, and the ring's physical gap is declared an imported kinematic finite-size gap. The nonlinearity and the coupling network are declared as imported boundary inputs; the retention hypothesis is flagged as an analogy across two AT systems, not a derivation. Both outcome branches are informative, and the deciding experiment costs one bench prototype.

1 Scope

What AT is. The Actualization Theory (AT) reconstructs physics on the canonical chain *Difference → Actualization → Inevitable Spectrum → Physics*, from two primitives — Difference (the fundamental boundary) and the tensor reference η — with the observable attractor realized on the 96-site circulant $C_{96}(1..6)$. The canonical monograph (v2.0, Ch1–Ch14) classifies every statement it makes as DERIVED, EMERGENT or BOUNDARY. Throughout this proposal the acronym "AT" denotes that framework only, and the phase labels AT-1xx / QGxxx denote its internal derivation and audit record.

This is a **proposal**. It contains no experimental result, changes no canonical AT claim, reclassifies no value, and introduces no primitive. Two ingredients are imported and are declared as boundary inputs: the locking nonlinearity (NP_005: the locking force is ABSENT, a boundary) and the engineered coupling network (NP_007/NP_011: the static coupling network is REFUTED as a physical field). The experiment can fail to falsify AT; it cannot confirm it.

2 The one idea worth doing

2.1 What is abandoned

Energy storage. A 96-node lattice stores microjoule-scale energy at kilowatt-scale overhead if cryogenic — some 10^6–10^9 times behind lithium-ion cells (~ 0.9 MJ kg^{-1}, ~ 5000 cycles, $\sim$100 EUR/kWh) on any storage figure of merit. No amount of AT machinery changes that, so the storage claim is closed here.

2.2 What survives

The lock lattice as a **phase-memory element**: a device whose state variable is *which modes are locked*, rather than how much charge is separated (capacitor) or how much chemical potential is stored (battery). Its retention hypothesis comes from the AT-120 one-way barrier — a topological charge, once actualized, cannot un-actualize — and its read-out from the AT-126 phase-interference law. We call it **LLPM** (Lock-Lattice Phase Memory):

> *An engineered circulant-$C_{96}(1..6)$ oscillator ring is charged into a locked mode configuration; the locked configuration is predicted to retain its state longer than a matched unlocked control, with the retention difference produced by the actualization barrier and readable through $\Delta\theta$ interference.*

The design is informative whether the answer is positive or negative, because AT's own Ch14 concedes that "standard complexity measures match or beat the lock rule on the evolving cohort": a statistical audit cannot settle the question. Only a physical phase transition can.

3 Three reasons

None of the three uses energy density as its metric.

3.1 Reason 1 — theory falsification (strongest)

Every existing lock-law result (QG307–QG319) is computed on datasets or synthetic cohorts. A hardware ring is the only place where Ch8 must produce a *dynamical phase transition* in a physical system instead of a statistic in a table.

Item	Setting
Metric	existence of a threshold in the *physical* coupling, with sharpness ≥ 3 and width ≤ 0.4 (the QG316 criticality criteria)
Bar	hysteresis area $A > 0$ at $\geq 5\sigma$ over ≥ 30 runs, absent in all three controls
If it passes	the lock structure is physically realizable; the lock law stops being a candidate descriptive statistic
If it fails	the lock law is demoted to a descriptive statistic (consistent with QG312's partial fake and QG319's 96.9% false-positive / 66.1% false-negative rates), and the storage claim is withdrawn permanently

Both branches are publication-grade information, at the cost of one bench prototype.

3.2 Reason 2 — phase / coherence memory

Here density is irrelevant; the metric is energy per operation: write energy, retention, read-out fidelity, variability. Because the state variable is *phase*, not charge, writing is phase-pinning

(M_002) rather than charge transfer, so the write-energy floor is set by phase noise rather than by Coulomb energy. The bar to matter is $< \sim 10$ fJ per write, retention comparable to a supercapacitor's self-discharge window, and variability $< 5\%$ (competitive with MRAM/PCM/FeRAM). The failure mode is thermal and technical phase diffusion: a room-temperature ring may retain nothing beyond $1/\omega_0 \times Q$. *Capacity caveat (post-fetch):* the D96 lattice is the *compressed* case, not the rich one — 5 survivors vs 17 for a random lattice (T_010), only 2 modes (1 distinct eigenvalue) within $2\lambda_2$ (T_014) — so D96 is chosen for derivability, **not** for state richness; a capacity-oriented device is the $\mathrm{D96}^3$ variant of the apparatus section.

3.3 Reason 3 — neuromorphic threshold element

The lock transition is a sigmoidal threshold with hysteresis — a synapse/neuron primitive. What is AT-specific is that the threshold position and sharpness are *predicted* by the operator and lock structure rather than fitted, which is precisely what phase-change and memristive devices cannot offer. The bar: beat PCM/memristor variability and switching energy at equal node count. The risk: the nonlinearity that produces locking is imported (NP_005) and may itself dominate the variability.

4 Canonical basis: the numbers the hardware must realize

The observable attractor is the circulant $C_{96}(1..6)$ (degree 12) with Laplacian spectrum

$$\lambda_k \;=\; 2\sum_{d=1}^{6}\Big(1-\cos\frac{2\pi dk}{96}\Big), \qquad k=0,\dots,95. \tag{1}$$

Equation (1) was recomputed independently for this proposal; the checks are listed in Table 1.

Quantity	Recomputed	Canonical	Check
λ_2 (first positive eigenvalue, $k=1$)	0.386351	0.3864	✓
$\omega_1 = \sqrt{\lambda_2}$	0.621571	0.6216	✓
$\omega_{\max}$ ($k=48$, $\lambda=12$)	3.979620	—	—
span $= \omega_{\max}/\omega_1$	6.402515	6.4025	✓
octave bands over the 95 positive modes	$[4,4,87]$	$[4,4,87]$	✓
zero modes	1	1	✓
mirror pairs $\lambda_k = \lambda_{96-k}$	47 pairs + self-conjugate $k=48$	47 pairs	✓
multiplicity multiset	$1+42\times 2+5+6=95$	$[42\times 2, 5, 6]$	✓
distinct eigenvalues	45	—	—
lock chain (D96-specific)	occMom$/\Sigma m = 20.0026$	20.0026	Ch8

Table 1: Verified spectrum data for $C_{96}(1..6)$. The canonical lock values are listed for reference only: by QG313 the lock *structure* is universal while the lock *values* are domain-specific, so a hardware ring must derive its own lock values and must not be scored against 20.0026.

Two canonical facts govern what hardware may claim. **(i) QG313:** lock structure is universal, lock values are domain-specific; the D96 values (20.0026, 2.4105, 8.2980) are therefore not hardware predictions. **(ii) QG316:** the critical transition $g^* \approx 0.31$ is a *synthetic-cohort* parameter — an exponent fraction over a 40-step organization ramp, disclosed as synthetic by MONO007 A08 — not a physical coupling, and it is not transferable. The hardware prediction is the *existence* of a threshold satisfying the criticality criteria, not the number 0.31. The inverse-spectral construction that converts a target spectrum into coupling weights already exists as an accepted ResearchY result (T_001, inverse DFT onto the first circulant row); no new mathematics is required to specify the hardware.

4.1 Post-fetch constraints on these numbers (rev. 2)

Three results that landed after the first draft constrain how the numbers above may be used.

1. The ring's gap is imported, not AT-derived (NP_169). For $C_N(1..6)$ the gap scales as $\lambda_{\text{gap}}(N) = 4\pi^2 \cdot 91/N^2 + O(N^{-4})$, so a positive limiting gap requires a scaling $f = \Theta(N^2)$, which NP_169 proves no AT-native primitive supplies: the only positive cases are the imported fixed-circumference length L_0 — a kinematic compact-domain gap — and a tautological density renormalization. At $N = 96$ the leading-order form gives 0.3898 against the *exact* 0.386351 used here (0.9%, i.e. the stated $O(N^{-4})$ term). Hence λ_2 must not be presented as a physical mass-gap scale in the hardware context, and at fixed circumference the gap degrades as $1/N^2$, which is a design input for any larger-N variant.

2. The near-gap band is essentially one doublet (T_014). Exact counting gives $N_{\text{gap}}(k) = \#\{\lambda \leq k\lambda_2\} =$ **2 modes (1 distinct eigenvalue)** at $k = 1.5, 2, 3$, and 4 modes (2 distinct) at $k = 4$. Only the fundamental doublet $k = \pm 1$ lies within twice the gap. Consequently the locking criterion must be defined per mode.

3. Robustness is degeneracy-controlled, and a structured lattice is not more robust (T_015). Under edge-removal, edge-addition and weight-noise perturbations D96 gives $\Delta\lambda_2/\lambda_2 = 0.0077$ with an attractor shift $\Delta A =$ **30** (the largest in the T_015 table, 44 degenerate levels), whereas the random lattice gives 0.0003 with $\Delta A = 0$. T_015 explicitly **refutes** "gap size predicts robustness", "near-gap density predicts robustness" and **"D96 outperforms random"** (H3). The retention comparison must therefore be controlled on the degeneracy class, since otherwise it confounds lock retention with degeneracy-driven spectral fragility.

5 Hypotheses and pass/fail criteria

6 Apparatus

6.1 Tier 1 (primary; bench electronic, days, low cost)

A 96-node ring of nonlinear oscillators with programmable circulant coupling.
Design constraints: node-to-node f_0 spread must be $< 1\%$ or the H0 test fails for trivial reasons — now an *anchored* tolerance, since T_015 measures $\Delta\lambda_2/\lambda_2 = 0.0077$ and $\Delta A = 30$ for D96 under weight-noise perturbations (44 degenerate levels splitting), so node and link-weight tolerances must be budgeted to keep the measured $\Delta\lambda_2/\lambda_2$ below the H0 gate, with ΔA recorded alongside (a degeneracy split is not visible as a pair-symmetry error); **no superiority claim** may be made, because T_015 refutes "D96 outperforms random"; phase noise sets the retention floor, since the locked/unlocked contrast is only measurable while $1/(Q\omega_0) \ll \tau_{\text{slow}}$; the nonlinearity strength is the analogue of the QG316 organization parameter, so the mapping must be pre-registered before data-taking and the raw control voltage reported alongside the normalized g.

6.2 Tier 2 (photonic chip, months)

96 coupled SiN microring resonators; the transmitted spectrum gives ω_k directly (H0), locking via the Kerr nonlinearity, read-out by phase-sensitive heterodyne detection. Costlier, better for H1, worse for H2 (loss).

Capacity-oriented variant (post-fetch). If state richness rather than derivability is the goal, T_012/T_013/T_014 point to the cubic tensor lattice $\mathbf{D96^3}$: 20,811 distinct non-zero eigenvalues, 16 survivors vs 5 for D96 (T_012), near-gap staircase 6/18/26/32 modes at $k =$

#	Hypothesis	Pass	Fail / interpretation
H0	the ring realizes $C_{96}(1..6)$	mirror-pair error $< 1\%$; bands $[4, 4, 87]$	absence = *validity* failure, not a theory failure
H1	a lock threshold exists in the physical coupling	sharpness ≥ 3 and width ≤ 0.4 in $f(g)$, hysteresis $A > 0$ at $\geq 5\sigma$	$f(g)$ smooth and reversible $\Rightarrow$ lock law demoted to descriptive statistic
H2	locked configurations retain state longer than controls	two time scales, $R = \tau_{\text{slow}}/\tau_{\text{fast}} \geq 10$, single-exponential in all controls	same decay law as controls $\Rightarrow$ the AT-120 analogy does not transfer; retention claim withdrawn
H3	read-out follows the interference law	$\lvert\Theta_{\text{mid}}\rvert \propto \cos(\Delta\phi/2)$, visibility $V \geq 0.90$	$V < 0.90$ or wrong functional form
H4	threshold element is competitive	beats PCM/memristor switching energy and variability	reason 3 retired; reasons 1–2 unaffected

Table 2: Hypotheses. H2 is flagged as the highest-risk, highest-information hypothesis: it is an analogy across two different AT systems (the R-field condensate of AT-120 and the circulant Laplacian ring of Ch6–Ch8), not a derivation. Testing an analogy is legitimate; asserting it as derived would not be. H2 also carries a known confound: T_015 shows a structured (degenerate) lattice is *more* spectrally fragile than a random one under symmetry-breaking, so a D96-vs-random retention comparison cannot by itself separate lock retention from degeneracy-driven fragility — hence the degeneracy-matched control below.

1.5/2/3/4 (T_014), compression ratio ≈1300 vs 8.8 (T_013). It is not the derivable spectrum (the canonical chain is 1D D96), and the legacy ∼19 species count is *not* recovered from it (T_011 refutes ∼19 from D96; T_012 gives 16, a ±5 correspondence, or 21 in the cubic sector) — so $D96^3$ is offered strictly as an engineering capacity path, not as a canonical object.

6.3 Tier 3 (superconducting, ∼1 year)

96 coupled superconducting cavities/transmons at $T < 20$ mK, $Q \approx 10^6$: the only tier in which a long retention claim is meaningful. Justified only if Tier 1 yields $A > 0$ at $\geq 5\sigma$, $R \geq 10$, and a device-level figure of merit beating a supercapacitor. Cryogenics are never justified by the energy stored.

7 Protocol

All runs use ≥ 30 independent repetitions per configuration, with the analysis script frozen before the first run.

R0 — calibration

measure each node's free-running ω_0 and Q; trim or calibrate to $< 1\%$ spread; record the spread as a systematic; additionally record the **degeneracy-split count** ΔA (the T_015 robustness measure — how many of the 44 distinct eigenvalues split under the board's residual symmetry-breaking).

R1 — validity (H0)

set w_d to the $C_{96}(1..6)$ values; measure ω_k by ring-down or spectral scan; compute the mirror-pair error and the octave band occupancy (expect $[4, 4, 87]$ over 95 positive modes). *Gate:* if this fails, stop — the apparatus is not a D96 lattice and nothing downstream is interpretable.

Element	Specification
Nodes	96 active resonators (LC tank + saturating amplifier), $f_0 = 100$ kHz–1 MHz
Nonlinearity	tanh-type saturation (Adler/Kuramoto phase dynamics) — *imported, BOUNDARY (NP_005)*
Coupling	for node i and each offset $d \in \{1..6\}$: links to $i \pm d$ via voltage-controlled transconductance $g_{m,d}$; 12 links per node
Programmability	one DAC per offset sets w_d; the Laplacian is then circulant with $\lambda_k = 2\sum_d w_d(1 - \cos 2\pi dk/96)$ — the same family as Eq. (1), with weights from T_001
Sweep	$g \equiv \sum_d w_d / \sum_d w_{d,\max}$, scanned over $g \in [0, 1]$ in 40 steps (mirroring the QG316 ramp)
Controls	(a) detuned ring (same Q, ω_k shifted out of ratio); (b) random-coupling ring (same Q and span); (c) linear-ramp spectrum (must fail CROWDING per QG305/QG308); (d) **degeneracy-matched** lattice - multiplicity spectrum matching D96's $\{1+42\times2+5+6\}$ (ΔA class) with different ω values, required by T_015 so that retention is not confounded with degeneracy-driven fragility
Read-out	96-channel multiplexed IQ demodulation at f_0, ≥ 1 MS/s per channel; θ_j and A_j per node
Injector	single-channel drive at selectable ω_k with programmable phase θ_{in}; abruptly removed for the retention run
Budget	2–4 $\times$ 16-channel ADC/DAC plus FPGA for real-time IQ: the EUR 5–20k class

R2 — charging and read-out (H3)
drive at ω_k to settling, then probe two adjacent sites with controlled relative phase $\Delta\phi$; measure $|\Theta_{\text{mid}}|/(|\Theta_{x_1}| + |\Theta_{x_2}|)$ versus $\Delta\phi$ and fit $\cos(\Delta\phi/2)$. *Gate:* $V \geq 0.90$.

R3 — lock threshold and hysteresis (H1)
sweep g upward over 40 steps and then downward; at each step, after settling, compute

$$f(g) = \frac{\#\{j : \ |d\Delta\theta_j/dt| < \varepsilon\}}{96}, \quad \varepsilon = 0.01\,\omega_1, \qquad A = \int |f_{\text{up}}(g) - f_{\text{down}}(g)|\,dg,$$

with sharpness = $\max|\Delta f|/\overline{|\Delta f|}$ and width = the $g_{10} \to g_{90}$ interval. Test sharpness ≥ 3, width ≤ 0.4, $A > 0$ at $\geq 5\sigma$. Because only the fundamental doublet lies within $2\lambda_2$ (T_014), $\varepsilon = 0.01\,\omega_1$ is a *lowest-doublet* criterion: define $\varepsilon_j = 0.01\,\omega_j$ per mode and report $f(g)$ both for the near-gap doublet and for the full 95-mode set, so a near-gap-only lock is not mistaken for a global one. Report the measured g_c as a hardware-specific value (QG313) and do *not* compare it with 0.31.

R4 — retention (H2)
(i) *ring-down*: charge into a locked configuration, remove the drive abruptly, record total stored energy $\sum_j A_j^2/2$ versus t for $\geq 10^4/\omega_1$; (ii) *cycle test*: $\geq 10^3$ charge/hold/discharge cycles, recording state fidelity after hold. Fit

$$\text{locked: } E(t) = a\,e^{-t/\tau_{\text{fast}}} + b\,e^{-t/\tau_{\text{slow}}}, \qquad \text{control: } E(t) = a\,e^{-t/\tau_c},$$

and report $R = \tau_{\text{slow}}/\tau_{\text{fast}}$ and the storage figure of merit FOM $= R \cdot E_{\text{ret}}/E_{\text{in}}$. *Pass:* $R \geq 10$ with the two-time-scale fit preferred by BIC/AIC against the single-exponential null, with all four controls single-exponential. **Mandatory degeneracy matching (post-fetch):** because T_015 shows the controlling factor is the degeneracy structure (ΔA), R4 is scored *separately* against control (d); a D96-vs-(b) advantage is not a result; each control's ΔA is reported beside its retention numbers; and if the advantage over (d) vanishes while the advantage over (b) persists, the effect is reported as a degeneracy/robustness artefact rather than as lock retention.

R5 — threshold benchmark (H4, optional)
only if R3 passes: measure switching energy per node, threshold sharpness and board-to-board variability across five identical boards; compare with published PCM/memristor figures.

R6 — controls and blinding
labels {locked, control a, b, c, d} are hidden from the analyst until the frozen script has produced numbers (QG319: FP 96.9%, FN 66.1% — precisely the failure mode to guard against). The lock classifier itself is fixed before R3.

8 Statistics, blinding and reproducibility

Rule	Setting
Repeats	≥ 30 per configuration; report mean $\pm$ 95% CI and effect size, never a single trace
Significance	$A > 0$ and $R \geq 10$ each at $\geq 5\sigma$
Model comparison	BIC and AIC for single- vs two-time-scale decay; the null is always the simpler model
Blinding	label blinding with a scripted or third-party unblinding step
Freezing	analysis-script hash recorded before R3; any change restarts the run
Negative controls	detuned, random-coupling, linear-ramp *and* **degeneracy-matched** lattices must fail H1 and H2 under the scoring rule of R4
Systematics	node-frequency spread, phase noise, ADC clock jitter, temperature drift and cable delay budgeted explicitly
Reproducibility	raw IQ traces archived with instrument settings and seeds

9 Kill criteria and decision tree

```
R1 fails            apparatus problem → repair and rerun (not a theory result)
R1 passes, R3 null      H1 falsified → withdraw the storage claim permanently
R3 passes, R4 null       keep the threshold-device direction (R3), drop phase
memory (R2)
R3+R4 pass, FOM < 10× supercap       stop; do not escalate on storage grounds
R3+R4 pass, FOM ≥ 10× supercap              escalate to Tier 2, then Tier 3
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Cryogenics are never justified by the energy stored.

10 Boundaries: what is imported

This proposal alters the DERIVED / EMERGENT / BOUNDARY status of no existing result.

11 What each outcome means

The negative branch is not a failure of the program: it is the honest closure of an open question, and it is the branch AT's own Ch14 anticipates.

Imported	Status	Consequence
locking nonlinearity	BOUNDARY (NP_005: locking force ABSENT)	the experiment tests the lock *structure*, not a derived interaction
engineered coupling network	BOUNDARY (NP_007/NP_011: REFUTED as a physical field)	the lattice is built, never claimed emergent
retention mechanism (AT-120 barrier)	analogy across systems	H2 is a hypothesis about transferability, not a derived prediction
$g^* = 0.31$	synthetic-cohort parameter (QG316; MONO007 A08)	a hardware g_c is expected to differ; quoting 0.31 as a hardware number would be a fit
D96 lock values 20.0026/2.4105/8.2980	D96-specific (QG313)	derive the hardware's own lock values from its own spectrum
the ring's physical gap	imported kinematic finite-size gap (NP_169)	the nonzero gap of a physical ring is the fixed-circumference case with an imported length L_0; AT supplies no natural scaling that preserves a positive gap as $N \to \infty$, so λ_2 is not a derived physical mass gap here
spectral superiority of the D96 lattice	REFUTED (T_015 H3)	do not claim the D96 lattice is more robust than the random or degeneracy-matched controls — it is *more* fragile ($\Delta A = 30$ vs 0)
capacity of the locked state set	EMERGENT / low for D96 (T_010/T_012/T_014)	report state richness as measured (D96 is the compressed case); $D96^3$ is a separate engineering device, not a canonical object
energy-storage claims	out of scope	the density argument is accepted as fatal

12 Deliverables

1. This proposal (PLANNED).
2. A numerical precursor (planned): a deterministic Kuramoto/Adler ring simulation on $C_{96}(1..6)$ (fixed seeds, no randomness) computing the expected sharpness/width curves and the two-time-scale signature, so that R3/R4 have an *a priori* prediction.
3. A canonical prediction-registry entry, quoted below, corresponding to **AT-P043** (AT-P042 being the tick discriminator, M_009). The canonical registry is *not* modified by this proposal.

```
\predstart{AT-P043}{PROPOSED}
\pfield{Title}{Lock-lattice physical realizability:  a lock threshold in an engineered C96
ring.}
\pfield{Derived From}{Difference → Actualization → Spectrum (D96) → Locking operator → lock
law (Ch8); criticality criteria from QG316; design recipe from ResearchY-T_001.}
\pfield{Prediction}{An engineered circulant C96(1..6) nonlinear oscillator ring exhibits a lock
threshold in the physical coupling (sharpness ≥ 3, width ≤ 0.4) with hysteresis, and a locked
configuration decays with two time scales against a single-exponential control.  Structure
universal (QG313); the threshold value and the lock values are hardware-specific and are NOT
predicted.}
\pfield{Experimental Test}{Bench electronic 96-node ring, runs R1-R4, ≥30 repeats, blinded
labels, four negative controls (detuned, random-coupling, linear-ramp, degeneracy-matched).}
\pfield{Current Evidence}{None -- proposal.  All existing lock-law evidence is statistical
(QG307-QG319).  Post-fetch inputs shaping the method:  NP_169 (the ring's gap is an imported
kinematic finite-size gap; no natural scaling preserves a positive gap), T_014 (only 2 modes
within 2λ₂), T_015 (robustness is degeneracy-controlled and a structured lattice is NOT more
robust than a random one).}
\pfield{Falsification Criterion}{No hysteresis, or locked retention decay indistinguishable
from all four controls.}
```

Outcome	Meaning for AT	Meaning for engineering
R1 only	the D96 lattice is realizable	an oscillator-ring testbed with a derivable spectrum
R1+R3	the lock structure is physically realizable, not merely statistical	a threshold device with predicted criticality criteria
R1+R3+R4	phase memory with barrier-protected retention	a candidate phase/coherence memory element (reason 2)
R1+R4 without R3	two-time-scale decay without a threshold — suspects an ordinary high-Q effect	no device claim; the lock reading is unsupported
R1 only, R3 null	lock law demoted to a descriptive statistic	the platform stays useful; the storage idea ends
any branch, capacity measured low	D96 is the compressed lattice (T_010/T_014) — no capacity claim is made	capacity reported as measured; $D96^3$ is the upgrade path
locked beats (b) but not (d)	degeneracy-driven fragility, not lock retention	report as an artefact (T_015); no device claim

13 Revision note (rev. 2): what the post-fetch results changed

The proposal was first written before the 2026-09-11 sync. The ten commits that landed (NP_169 and T_007–T_015) were checked against every claim here. No result invalidates the experiment and none reclassifies a canonical object, but three change the method and two the framing; none of those commits touched the prediction registry, the classification registry or the coverage record, so the drafted AT-P043 remains unclaimed and the QG313/QG316 citations remain current.

Post-fetch result	Effect	Where
T_015: robustness is degeneracy-controlled; "D96 outperforms random" REFUTED (D96 $\Delta\lambda_2/\lambda_2 = 0.0077$, $\Delta A = 30$; random 0.0003, $\Delta A = 0$)	**Method change** — mandatory degeneracy-matched control (d); R4 scored against it; ΔA reported per control; a D96-vs-random advantage is not a result; $< 1\%$ spread becomes an anchored tolerance; no superiority claim	apparatus, H2, R0/R4/R6, statistics, boundaries, outcomes
T_010/T_012/T_014: D96 is the compressed, low-capacity lattice (5 survivors vs 17; 2 modes within $2\lambda_2$; $D96^3$ 16)	**Framing change** — D96 chosen for derivability, not capacity; capacity caveat in reason 2; $D96^3$ as an explicitly non-canonical engineering variant; legacy $\sim$19 not used	reasons, apparatus, boundaries, outcomes
NP_169: no AT-native scaling preserves a positive mass gap ($\lambda_{\text{gap}} = 4\pi^2 \cdot 91/N^2 + O(N^{-4})$)	**Framing change** — the ring's gap is an imported kinematic finite-size gap; λ_2 is not a physical mass-gap scale here; the $1/N^2$ gap penalty is a design input for larger N	canonical basis, boundaries
T_014: near-gap density is exact counting (2 modes / 1 distinct at $k \leq 3$)	**Protocol refinement** — per-mode $\varepsilon_j = 0.01\omega_j$; near-gap and global lock fractions reported separately	canonical basis, R3
T_007–T_009, T_011, T_013	No change (population dynamics on spectral landscapes; compression attributed to landscape size; $\sim$19 refuted from D96 but never used here)	—

Unchanged by the revision: the core experiment (R1–R5), the three reasons, the kill criteria, the density argument, the AT-P043 draft, and every verified D96 number ($\lambda_2 = 0.386351$, $\omega_1 = 0.621571$, span $= 6.402515$, bands $[4, 4, 87]$, 47 mirror pairs, multiplicities $\{2 \times 42, 5, 6\}$).

14 Conclusion

One idea survives the density objection: the lock lattice as a phase-memory and threshold element whose distinguishing feature is a predicted criticality, tested on hardware for the first time. Three reasons justify it — falsification of the lock law as a dynamical claim (strongest), phase memory where density is irrelevant, and a neuromorphic threshold element with predicted sharpness. The deciding experiment costs one bench prototype, and both branches of the outcome are informative. Energy storage is explicitly abandoned, the nonlinearity and coupling network are explicitly imported, and the retention hypothesis is explicitly an analogy awaiting test rather than a derivation.

Status: PROPOSAL, rev. 2 — no experiment performed, no result claimed, no canonical AT statement modified.